

The Ising gap gives a conditional local parent Hamiltonian

The square-root Gibbs transform of C124 is a positive, three-site local, frustration-free Hermitian operator with unique ground state $\sqrt{\pi_{N,b}}$ and exact gap $a[1 - \tanh(2b)]$. Multiplication by a supplied action constant gives an energy-unit Hamiltonian. This construction preserves the chosen stochastic clock and does not select physical dynamics or action units. It resolves G04's finite-volume transfer decision.

1. Object, domain and transform

Use C124's model: $N \geq 3$ periodic spins, $0 \leq b < \infty$, per-site refresh rate $a > 0$ and stationary law $\pi(\sigma) = Z^{-1} \exp(b \sum_i \sigma_i \sigma_{i+1})$. The generator $Q = a \sum_i (E_i - I)$ acts on functions in complex $L^2(\pi)$; E_i is conditional expectation over spin i . Its time parameter is the auxiliary sampling clock and a has inverse-time units.

Let $D = \text{diag } \pi$ and $U : L^2(\pi) \rightarrow \mathcal{H}_N = (\mathbb{C}^2)^{\otimes N}$ be $(Uf)_\sigma = \sqrt{\pi(\sigma)} f(\sigma)$. The codomain has the standard counting inner product; U is unitary between these two inner-product spaces. All operators below have the entire finite-dimensional space as domain. Set

$$A_N = -UQU^{-1} = -D^{1/2}QD^{-1/2}.$$

Detailed balance makes this matrix Hermitian. Its quadratic form is

$$\langle \psi, A_N \psi \rangle = \frac{1}{2} \sum_{\sigma, \eta} \pi(\sigma) Q_{\sigma\eta} \left| \frac{\psi_\eta}{\sqrt{\pi(\eta)}} - \frac{\psi_\sigma}{\sqrt{\pi(\sigma)}} \right|^2 \geq 0,$$

where diagonal terms vanish. Irreducibility implies a one-dimensional kernel, spanned by the normalized vector

$$|\Omega\rangle = Z^{-1/2} \sum_{\sigma} e^{(b/2) \sum_i \sigma_i \sigma_{i+1}} |\sigma\rangle.$$

Its squared basis amplitudes reproduce the Gibbs probabilities. Regarding those amplitudes and other vectors as physical quantum states requires a state/measurement interpretation in addition to this Hilbert-space map.

2. Explicit local terms and their positivity

Write $Z_i|\sigma\rangle = \sigma_i|\sigma\rangle$ and let X_i flip spin i . Put $\theta = \tanh(2b)$, $s = \text{sech}(2b)$, $c_+ = (1 + s)/2$, $c_- = (s - 1)/2$. Then

$$A_N = \sum_i h_i, \quad h_i = \frac{a}{2} \left[I - \frac{\theta}{2} Z_i (Z_{i-1} + Z_{i+1}) - (c_+ I + c_- Z_{i-1} Z_{i+1}) X_i \right].$$

Derivation. For fixed neighbours with sum $S = \sigma_{i-1} + \sigma_{i+1}$, the original flip rate is $r_i(\sigma) = a[1 - \sigma_i \tanh(bS)]/2$. Detailed balance gives transformed off-diagonal entry $-\sqrt{r_i(\sigma)r_i(\sigma^i)} = -a/[2 \cosh(bS)]$. The diagonal entry is unchanged. Since $S = 0, \pm 2$, the off-diagonal coefficient equals $c_+ + c_- \sigma_{i-1} \sigma_{i+1}$, giving the formula. The two neighbours are distinct even for $N = 3$; all indices are cyclic.

Alternatively $h_i = a(I - UE_iU^{-1})$. Each E_i is an orthogonal projection, so $h_i \geq 0$, $h_i^2 = ah_i$ and $h_i|\Omega\rangle = 0$. Thus the sum is frustration-free, although its overlapping terms need not commute. Each term has norm a and support in the three sites $i-1, i, i+1$; the global density transform has not introduced interactions of growing range. For a fixed neighbour block the local ground vector has components proportional to $e^{bS/2}, e^{-bS/2}$, which also checks the square-root and sign conventions. At $b = 0$, $h_i = a(I - X_i)/2$ and the ground state is the product of plus- X vectors. These statements are direct finite-dimensional derivations.

3. Exact gap, normalization and limits (C126)

Unitary equivalence and C124 give

$$\text{gap}(A_N) = a[1 - \tanh(2b)].$$

The corresponding nonzero eigenvector is UM with $M(\sigma) = \sum_i \sigma_i$, orthogonal to $|\Omega\rangle$ by spin-flip symmetry. This is an all-mode spectral statement, inherited from C124 rather than inferred from locality or the ground state alone.

For a supplied constant $K > 0$ with action units, define $H_N = KA_N$. Its ground energy is zero and its excitation gap in energy units is

$$\Delta_N = Ka[1 - \tanh(2b)].$$

For $K_N \geq K_0 > 0$, $a_N \geq a_0 > 0$, $0 \leq b_N \leq B < \infty$, $\Delta_N \geq K_0 a_0 [1 - \tanh(2B)]$ uniformly in N . These are bounds on a sequence of finite-volume Hamiltonians. They do not construct an infinite-volume Hilbert representation or establish a continuum gap. Fixed total refresh rate replaces a by a/N and closes the gap. Growing b_N or vanishing K_N can close it as well when the other parameters stay fixed.

4. Which physical premise has been supplied?

The exact semigroup identity is

$$e^{-tH_N/K} = Ue^{tQ}U^{-1}.$$

It identifies the transformed sampling semigroup with imaginary-time evolution of the constructed operator, in the same time units. Defining the unitary $e^{-itH_N/K}$ is mathematically possible but is an additional choice of real-time evolution; it is not the original stochastic transition law.

Input	Use	What remains to identify physically
Positive Gibbs law and detailed balance	Unitary weighted-space transform and Hermiticity	Physical state and observable interpretation
Single-site heat-bath generator	Three-site terms and exact inherited gap	Why this generator governs physical imaginary time
Per-site rate a	Overall spectral rate scale	Conversion from sampling time to physical time
Supplied action constant K	Converts inverse time to energy and sets phase units	Positive universal action normalization
Uniform K, a, b bounds	Finite-volume energy-gap lower bound	Their physical origin and any further limit construction

The classical energy function $-J \sum_i \sigma_i \sigma_{i+1}$ used to prepare π is not the constructed H_N : the latter has off-diagonal spin flips and coefficients depending on $b = \beta J$ and Ka . Knowledge of J and β fixes neither the refresh clock nor a universal action constant. For the same π , replacing Q by λQ , $\lambda > 0$, preserves the ground state and locality while multiplying every excitation gap by λ . Holding Q fixed and varying K independently gives the other normalization freedom. Thus the map yields a conditional stochastic-parent Hamiltonian; an independent physical identification is still a premise.

5. Evidence and strategic consequence

Written proof and literature status are recorded separately in the ledger. The B71 audit matches the established detailed-balance parent mapping in Henley and Castelnovo et al.; the explicit Ising term is a derived specialization. The written review checks its coefficients and inherited gap. No novelty is asserted.

G04 stops at the explicit finite-volume operator. Additional Ising spectra would not select its clock or action scale. A further operator task requires an independently specified physical dynamics whose imaginary-time generator can be compared with this one. The quantum track's reversible-descent premise is selected next: test whether a specified interacting mechanical orientation flow closes on C125's moment coordinates. STATE owns that bounded decision.